



DEPARTMENT OF MATHEMATICS

Academic Year 2025-2026 (Even Semester)

Date: 17 April 2026	CIE-I	Max Marks: 10+50
Time: 2:00 PM to 4:00 PM	UG	Duration: 120 minutes
Semester: II, B. E. (BT, CD, CI, CS, CY)		
Course Title: Number Theory Vector Calculus and Computational Methods		Course Code: MA221TC

Q. No.	SCHEME AND SOLUTIONS								M	
Q1	$\Delta^3 f(x) = -385(3!)(2)^3 = -18480, \Delta^5 f(x) = 0$								1+1	
Q2	$x = \frac{(y-y_1)(y-y_2)}{(y_0-y_1)(y_0-y_2)}x_0 + \frac{(y-y_0)(y-y_2)}{(y_1-y_0)(y_1-y_2)}x_1 + \frac{(y-y_0)(y-y_1)}{(y_2-y_0)(y_2-y_1)}x_2 = 1$								1+1	
Q3	$17 = gcd(a, b) = gcd(a, b + ay) = gcd(6409, 42823 + 6409y) = gcd(6409, 49232)$								1+1	
Q4	<i>if $p = 2$ then $p + 2 = 4$ is not a prime and if $p = 3$ then $p + 2 = 5$ is a prime</i>								1+1	
Q5	$999 = 3^3 \times 37 \therefore \tau(999) = (3 + 1)(1 + 1) = 8$.								1+1	
T1	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$p = \frac{x - x_0}{h} = \frac{7 - 0}{8} = 0.875$ $O(7) = 12.139$ $p = \frac{x - x_n}{h} = \frac{27 - 40}{8} = -1.625$ $O(27) = 7.9682$ $\Delta^2(O(24)) = 0.221$.		4	
	14.621	-2.778	0.805						1	
	11.843	-1.973	0.521	-0.284					2	
	9.87	-1.452	0.339	-0.182	0.102				1	
	8.418	-1.113	0.221	-0.118	0.064	-0.038			2	
	7.305	-0.892								
	6.413									
T2a	$f(x) = \frac{(x-2)(x-3)(x-5)}{(1-2)(1-3)(1-5)} \times 3 + \frac{(x-1)(x-3)(x-5)}{(2-1)(2-3)(2-5)} \times 6 + \frac{(x-1)(x-2)(x-5)}{(3-1)(3-2)(3-5)} \times 19$ $+ \frac{(x-1)(x-2)(x-3)}{(5-1)(5-2)(5-3)} \times 99 = \mathbf{x^3 - x^2 - x + 4}$								3	
	$f(4) = 48$								2	
									1	
T2b	x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$h = 3, \left[\frac{dy}{dx} \right]_{x=x_n=9} = \frac{1}{h} \left(\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right)$ $= 2.5222$ $\left[\frac{d^2 y}{dx^2} \right]_{x=x_n=9} = \frac{1}{h^2} \left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n \right) = -0.3566$			2	
	0	0								
	3	1	1	6	-4.6					
	6	8	7	1.4						
	9	16.4	8.4							
T3a	x	0	0.5	1.0	1.5	2.0	2.5	3.0	2	
	f(x)	0.3989	0.3521	0.2420	0.1295	0.0540	0.0175	0.0044		
	$h = 0.5$									
	(i)	Simpson's 3/8 rule								
		$\int_0^3 f(x) dx = \frac{3}{8} h(y_0 + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5) + y_6) = 0.4986$							2	
T3a	(ii)	Weddle's rule								
		$\int_0^3 f(x) dx = \frac{3}{10} h(y_0 + y_2 + 6y_3 + y_4 + 5(y_1 + y_5) + y_6) = 0.4987$							2	

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T3b	$34x \equiv 60 \pmod{98}$. By Euclid's algorithm $(34, 98) = 2$ and $2 \nmid 60 \Rightarrow 2$ incongruent solutions exist. $34x \equiv 60 \pmod{98} \Rightarrow 17x \equiv 30 \pmod{49} \Rightarrow 17x + 49y = 30$. By Euclid's algorithm $15 = 49 + 17(-2), \quad 2 = 17 + 15(-1), \quad 1 = 15 + 2(-7)$ By back substitution we get $1 = 17(-23) + 49(8) \Rightarrow 30 = 17(-690) + 49(240)$ $\therefore x \equiv -690 \pmod{49} \equiv 45 \pmod{49}$ $\therefore x \equiv 45 \pmod{98}$ and $x \equiv 94 \pmod{98}$	1
		2
		1
T4a	By Euclid's algorithm $12378 = 3054(4) + 162, \quad 3054 = 162(18) + 138, \quad 162 = 138 + 24$ $24 = 18 + 6, \quad 18 = 6(3) + 0 \Rightarrow (12378, 3054) = 6$. By back substitution $6 = \gcd(12378, 3054) = 12378(132) + 3054(-535)$	2
		1
		2
T4b	The last two digits of $7777^{3333} = 7777^{3333} \pmod{100}$. $(7777, 100) = 1$ and $\phi(100) = \phi(2^2 \times 5^2) = 100 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 40$ $\therefore 7777^{\phi(100)} \equiv 1 \pmod{100} \Rightarrow 7777^{40} \equiv 1 \pmod{100} \Rightarrow (7777^{40})^{83} \equiv 1^{83} \pmod{100}$ $\Rightarrow 7777^{3320} \equiv 1 \pmod{100}$ Also $7777^{12} \equiv 21 \pmod{100}$ and $7777^{13} \equiv 17 \pmod{100}$ $\Rightarrow 7777^{3333} \equiv 17 \pmod{100}$	1+1
		1
		1
		1
T5	$x \equiv 5 \pmod{6}, \quad x \equiv 4 \pmod{11}, \quad x \equiv 3 \pmod{17}$ $a_1 = 5, a_2 = 4, a_3 = 3, m_1 = 6, m_2 = 11, m_3 = 17$ & 6, 11, 17 are pair wise coprime $\Rightarrow$ Unique Soln. exists	1
	$M = m_1 m_2 m_3 = 1122, M_1 = \frac{M}{m_1} = 187, M_2 = \frac{M}{m_2} = 102, M_3 = \frac{M}{m_3} = 66$	1
	$M_1 M_1^{-1} \equiv 1 \pmod{m_1} \Rightarrow 187 M_1^{-1} \equiv 1 \pmod{6} \Rightarrow M_1^{-1} = 1$	2
	$M_2 M_2^{-1} \equiv 1 \pmod{m_2} \Rightarrow 102 M_2^{-1} \equiv 1 \pmod{11} \Rightarrow M_2^{-1} = 4$	2
	$M_3 M_3^{-1} \equiv 1 \pmod{m_3} \Rightarrow 66 M_3^{-1} \equiv 1 \pmod{17} \Rightarrow M_3^{-1} = 8$	2
	$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + a_3 M_3 M_3^{-1}) \pmod{M} \equiv 4151 \pmod{1122} \equiv 785 \pmod{1122}$	2